Virtual Substitution Scan Via Single-Step Free Energy Perturbation

Ying-Chih Chiang, Yi Wang
Department of Physics, the Chinese University of Hong Kong, Shatin, N.T., Hong Kong

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ABSTRACT:
With the rapid expansion of our computing power, molecular dynamics (MD) simulations ranging from hundreds of nanoseconds to microseconds or even milliseconds have become increasingly common. The majority of these long trajectories are obtained from plain (vanilla) MD simulations, where no enhanced sampling or free energy calculation method is employed. To promote the "recycling" of these trajectories, we developed the Virtual Substitution Scan (VSS) toolkit as a plugin of the open-source visualization and analysis software VMD. Based on the single-step free energy perturbation (sFEP) method, VSS enables the user to post-process a vanilla MD trajectory for a fast free energy scan of substituting aryl hydrogens by small functional groups. Dihedrals of the functional groups are sampled explicitly in VSS, which improves the performance of the calculation and is found particularly important for certain groups. As a proof-of-concept demonstration, we employ VSS to compute the solvation free energy change upon substituting the hydrogen of a benzene molecule by 12 small functional groups frequently considered in lead optimization. Additionally, VSS is used to compute the relative binding free energy of four selected ligands of the T4 lysozyme. Overall, the computational cost of VSS is only a fraction of the corresponding multi-step FEP (mFEP) calculation, while its results agree reasonably well with those of mFEP, indicating that VSS offers a promising tool for rapid free energy scan of small functional group substitutions. © 2016 Wiley Periodicals, Inc. Biopolymers 105: 324–336, 2016.

Keywords: *molecular dynamics; free energy calculation; single step FEP; lead optimization*

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Correspondence to: Y. Wang; e-mail: yiwang@phy.cuhk.edu.hk
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INTRODUCTION

The computation of free energy is central to the quantitative characterization of many biological processes,[1–7] including protein-ligand binding,[8–10] protein-protein recognition,[11,12] solvation of small molecules[13,14] as well as their permeation through lipid membranes.[15,16] In the context of protein-ligand binding, such computation can be employed to obtain the absolute affinity of a given ligand as well as the relative binding strength of two ligands.[17–19] Calculation of the latter finds an important application in computer-aided drug design[20]—by predicting whether a certain chemical modification will enhance the activity of a lead compound, one can better plan the subsequent synthesis of new compounds. For example, a "chlorine scan" guided by free energy perturbation (FEP) was used to identify the most promising sites for substitution of aryl hydrogens, which, along with other calculations, led to the optimization of an initially inactive azole compound to an anti-HIV agent with lower nanomolar activity.[21,22]

The accurate and efficient evaluation of free energy is often hindered by issues such as limited sampling, force field inaccuracy or insufficient coverage of chemical space, as well as errors incurred during the setup and analysis of the calculation.[23–26] In recent years, considerable efforts have been made to address these issues[6,19,27–29] and toolkits designed to automate the preparation, setup and analysis of free energy calculation have been developed.[30–32] Accompanying this methodological development is the rapid expansion of our computing power. With high performance clusters and special-purpose machines, simulations up to hundreds of nanoseconds are performed routinely, and reports of microsecond to millisecond simulations are no longer uncommon.[33,34] While free energy calculation can directly benefit from such an increase in sampling time, most of the reported long trajectories are the so-called "vanilla" Molecular Dynamics (MD) simulations, i.e., they do not employ any specially constructed Hamiltonian, enhanced sampling scheme, or free energy calculation technique.

To promote the post-processing of vanilla MD trajectories for free energy calculations, we develop Virtual Substitution Scan (VSS) toolkit, which utilizes the single-step free energy perturbation (sFEP) theory[35] to post-process a trajectory for a quick FEP scan. The sFEP scheme, in combination with a soft-core potential,[36,37] i.e. a modified Lennard-Jones potential, has been applied to determine the solvation free energy for nonpolar solutes,[38] protein-ligand binding free energy,[39–41] as well as the solvation free energy for polar solutes with additional sampling on molecular rotation and translation.[42] Using an unphysical reference structure constructed with soft-core potential, sFEP calculation has also been performed on structurally diverse compounds.[43] Alternatively, one may apply the envelope distribution sampling (EDS) method to construct a reference state through combining Hamiltonians of multiple end states.[44–46] While the above studies clearly demonstrate the usefulness of a carefully crafted Hamiltonian, for relatively simple substitutions (halogens, hydroxyl, and methyl), a good agreement between sFEP calculation and experimental results has been achieved based on a single, vanilla MD trajectory.[47] Here, we employ VSS to investigate a larger sets of substitutions via sFEP calculation based on vanilla MD trajectories. Following the implementation of VSS as a plugin for the visualization and analysis program VMD,[48] we apply it to compute the free energy change upon substituting aryl hydrogens by 12 small functional groups frequently encountered in the study of protein-ligand binding and lead optimization[21,22,49,50]: -Cl, -CH_3, -NH_2, -OCH_3, -CN, -CHO, -CH_2OH, -CH_2NH_2, -$NHCH_3$, -OH, -F, and -CH_2CH_3. Unlike a previous study,[47] we sample explicitly dihedral rotations of the substituting groups, since such rotations are generally accessible under room temperature.[38] Two test systems in aqueous solution, namely, a free benzene and a benzene-T4 Lysozyme complex, are employed. All VSS results are compared with multi-step FEP (mFEP) calculations carried out with the program NAMD,[51] under the same force field and simulation conditions. This comparison provides direct evaluation of VSS performance, regardless of the force field or simulation parameters employed in the current work.

The rest of the article is organized as the following: the method section contains the derivation of the VSS working equation, implementation, and simulation protocols. The results of free energy calculations, obtained with both NAMD and VSS, are then discussed. This discussion is followed by detailed error analysis via the subsampling method[26,52–55] which sheds light onto the precision and accuracy of the VSS calculation. Finally, a brief conclusion is provided in the last section.

METHODS

In this section, we will first briefly review the FEP equation and then derive the VSS working equation. For convenience, the derivation is based on the canonical ensemble, while the adaptation to the isothermal-isobaric ensemble is straightforward. Implementation details of VSS and the protocols of simulations are then provided.

THEORY

The relative binding (or solvation) free energy between two ligands L_1 and L_2 can be obtained via evaluating the free energy changes of two alchemical transformation processes in a thermodynamic cycle.[3] For convenience, we consider the alchemical transformation of a free ligand L_1 into L_2, in aqueous solution. The former, L_1 in aqueous solution, is often termed the reference state (R), while the latter is called the target state (T). The free energy change ΔA can be evaluated by the forward sFEP equation[35]:

$$e^{-\beta\Delta A} = \frac{Z_T}{Z_R} = \langle e^{-\beta u} \rangle_R\,, \tag{1}$$

where $\beta = 1/k_B T$, with k_B representing the Boltzmann constant and T the absolute temperature, Z_i denotes the partition function of state i ($i = T,R$), and u is the potential energy difference between the target state and the reference state. The corresponding backward sFEP equation reads:

$$e^{\beta\Delta A} = \frac{Z_R}{Z_T} = \langle e^{\beta u} \rangle_T\,. \tag{2}$$

The above transition between the reference and target states can be achieved through multiple intermediate states, resulting in the multi-step FEP (mFEP) scheme.[3,4,6]

Our goal is to reformulate Eq. (1) to compute the free energy change based on a vanilla MD trajectory with only one ligand, namely L_1. As Eq. (1) implies that the total number of particles is conserved during the alchemical transformation and since L_1 and L_2 may have different numbers of atoms, in standard MD packages, a complementary "ghost" ligand is often introduced in both the reference and the target states. Such a strategy is known as the dual-topology setting[56] and has been employed by the program NAMD,[51] which treats the ligands with soft-core potential.[36,37] Within the dual-topology setting, the added ghost ligand does not interact with the rest of the system. For instance, when L_2 is added as a ghost to the aqueous solution of L_1, there is no coupling between L_2 and L_1, nor between L_2 and water. Besides, the ghost ligand is often treated as an "ideal gas molecule" whose intra-molecular bonded interaction is kept.[56] Following this setting, the potential energy function of the reference state reads,

$$U_R(\mathbf{r_0},\mathbf{r_1},\mathbf{r_2})=U'_R(\mathbf{r_0},\mathbf{r_1})+U_2^{bonded}(\mathbf{r_2})\,, \tag{3}$$

where $\mathbf{r_0}$ represents coordinates of the part common to both states (water), $\mathbf{r_1}$ and $\mathbf{r_2}$ are coordinates of L_1 and L_2, respectively. The term $U_2^{bonded}(\mathbf{r_2})$ represents the bonded energy of the ghost ligand L_2, while $U'_R(\mathbf{r_0},\mathbf{r_1})$ denotes the energy of a system with only L_1 in solution, the latter of which corresponds to the system used in a vanilla MD simulation:

$$\begin{aligned} U'_R(\mathbf{r_0},\mathbf{r_1}) &= U_0^{bonded}(\mathbf{r_0})+U_1^{bonded}(\mathbf{r_1}) \\ &+U_0^{nonb}(\mathbf{r_0})+U_1^{nonb}(\mathbf{r_1})+U_{0,1}^{nonb}(\mathbf{r_0},\mathbf{r_1})\,. \end{aligned} \tag{4}$$

In the above equation $U_i^{bonded}(\mathbf{r_i})$ and $U_i^{nonb}(\mathbf{r_i})$ denote the bonded and nonbonded energy of the part with coordinate $\mathbf{r_i}$ ($i=0, 1$), and $U_{0,1}^{nonb}(\mathbf{r_0},\mathbf{r_1})$ represents non-bonded interaction energy between particles belonging to $\mathbf{r_0}$ and particles belonging to $\mathbf{r_1}$. Similarly, a ghost ligand L_1 is introduced into the target state:

$$U_T(\mathbf{r_0},\mathbf{r_1},\mathbf{r_2})=U'_T(\mathbf{r_0},\mathbf{r_2})+U_1^{bonded}(\mathbf{r_1})\,, \tag{5}$$

where $U'_T(\mathbf{r_0},\mathbf{r_2})$ denotes the potential energy of a system with only L_2 in solution:

$$\begin{aligned} U'_T(\mathbf{r_0},\mathbf{r_2}) &= U_0^{bonded}(\mathbf{r_0})+U_2^{bonded}(\mathbf{r_2})+U_0^{nonb}(\mathbf{r_0}) \\ &+U_2^{nonb}(\mathbf{r_2})+U_{0,2}^{nonb}(\mathbf{r_0},\mathbf{r_2})\,. \end{aligned} \tag{6}$$

We now reformulate Eq. (1) by separating the ghost ligand's partition function after inserting Eqs. (3–6) to Eq. (1):

$$e^{-\beta\Delta A}=\frac{Z_T}{Z_R}=\frac{\int d\mathbf{r_0}\int d\mathbf{r_1}\, e^{-\beta U'_R(\mathbf{r_0},\mathbf{r_1})}\,\frac{\int d\mathbf{r_2} e^{-\beta U_2^{bonded}(\mathbf{r_2})}e^{-\beta u}}{\int d\mathbf{r_2} e^{-\beta U_2^{bonded}(\mathbf{r_2})}}}{\int d\mathbf{r_0}\int d\mathbf{r_1}\, e^{-\beta U'_R(\mathbf{r_0},\mathbf{r_1})}}=\langle\langle e^{-\beta u}\rangle_{\substack{bonded\\2}}\rangle_{R'}\,, \tag{7}$$

where the subscripts $\substack{bonded\\2}$ and R' indicate ensembles governed by the potential energy functions $U_2^{bonded}(\mathbf{r_2})$ and $U'_R(\mathbf{r_0},\mathbf{r_1})$, respectively, and the potential energy difference u now reads,

$$u=U_2^{nonb}(\mathbf{r_2})+U_{0,2}^{nonb}(\mathbf{r_0},\mathbf{r_2})-U_1^{nonb}(\mathbf{r_1})-U_{0,1}^{nonb}(\mathbf{r_0},\mathbf{r_1})\,. \tag{8}$$

Equation (7) is the working equation of VSS, which expresses $e^{-\beta\Delta A}$ as a double ensemble average: the outer layer is the ensemble average of system R', collected from a MD simulation with only L_1 in solution, while the inner layer is the ensemble average of the ghost ligand L_2, generated by VSS, according to the Boltzmann factor $e^{-\beta U_2^{bonded}}$. Clearly, only L_2 configurations with non-negligible Boltzmann factors contribute significantly to the ensemble average $\langle\ldots\rangle_{\substack{bonded\\2}}$. This excludes most configurations where bond length or angles deviate from their equilibrium values, as these degrees of freedom (DOFs) are usually governed by steep potential energy functions. Dihedral rotations, on the contrary, are often accessible under room temperature, due to relatively low energy barriers on their potential surfaces. For the above reason, the dihedral rotation of the substituting functional group is sampled explicitly during the VSS calculation, i.e., each desired dihedral is rotated at equal spacing and the resulting u and $U_2^{bonded}(\mathbf{r_2})$ are collected for the evaluation of Eq. (7). Such an explicit sampling of dihedrals distinguishes VSS from some of the previous post-processing methods,[47] and is found particularly important for certain functional groups (see Results section).

IMPLEMENTATION

VSS is implemented as a plugin of the visualization and analysis program VMD.[57] It evaluates the free energy change upon mutating L_1 into L_2 via Eq. (7). The computation of the potential energy terms in Eq. (7), e.g., u and $U_2^{bonded}(\mathbf{r_2})$, closely follows the standard packages. Specifically, computation of the vdW interactions employs 1-4 scaling exclusion and a cutoff scheme with a switching function as implemented in NAMD,[51] while the electrostatic energy is computed via the Particle Mesh Ewald (PME) algorithm[58] as implemented in the PMEPOT plugin by Aksimentiev and Schulten.[59] To further speed up the calculation, space decomposition technique, and parallelization with OpenMP are both employed. Following the free energy calculation, an optional subsampling calculation can be performed according to Ytreberg and Zuckerman,[55] which provides an estimate of uncertainty in the reported ΔA (see Simulation protocols).

When performing the calculation, VSS processes the trajectory frame by frame. For each frame of a trajectory containing L_1, VSS generates an ensemble for ligand L_2, as required by Eq. (7). This ensemble is collected by rotating the dihedral angles of the functional groups of L_2, whose core structure is taken from the structure of L_1 at a given frame, and the substituting functional group (geometry pre-minimized based on $U_2^{bonded}(\mathbf{r_2})$) then replaces the original functional group in L_1, e.g., $-CH_3$ replacing -H on a benzene. We refer to this treatment as "on-site mutation," since positions of all L_2 atoms (except for the substituting functional group) are kept the same as their L_1 counterparts. According to its code design, VSS is similar to a dual-topology method, viz. both L_1 and L_2 have their own coordinates independently stored in the computer's memory. Yet, our on-site mutation strategy makes the method more single-topology-like in practice, viz. the common part of L_1 and L_2 have identical positions. To avoid any confusion, we use only the term "on-site mutation" to refer to our topology setup.

The above protocol implies that for a given frame of the post-processed trajectory, the six external DOFs involving translation and rotation of L_2 is fixed. This is equivalent to the protocol used by NAMD when L_1 and L_2 share a large enough common part. In fact, for NAMD FEP calculation, even if the common part of the two molecules is not large enough, e.g., consisting of zero to two atoms, restraints are desirable to ensure that L_2 does not drift away from the position or orientation of L_1. Otherwise, one may encounter sampling issues akin to the "wandering ligand" problem in the calculation of absolute binding free energy.[17] One such case will be discussed in the results section.

To use VSS, the user needs to provide the trajectory and psf files of L_1, the pdb and psf files of L_2, as well as the parameter file to be used. Currently, the plugin reads only the dcd trajectory and the CHARMM force field parameters. Different force fields and trajectory formats will be supported in the future. Detailed settings are provided as a Tcl script together with the package and its tutorial, which will be available through the authors' website. The plugin writes a log file (vss.log) for the free energy change computed over the trajectory, along with the effective u [see Eq. (7)] for each frame. The error analysis can be performed by the user after the calculation, using the standard subsampling package[26,53,55] or an analysis routine adapted by us.

SIMULATION PROTOCOLS

Our first test system consists of a free benzene in water, which is alchemically transformed into various derivatives by substituting an aryl hydrogen with different functional groups. The twelve chosen functional groups (Figure 1), which have different polarity and sizes, are often considered in lead optimization.[21,22] VSS calculations of aryl hydrogen substitution by these groups are performed on a 50-ns trajectory of a benzene molecule in water (box size 30 Å by 30 Å by 30 Å). For each polyatomic functional group, dihedral conformations are

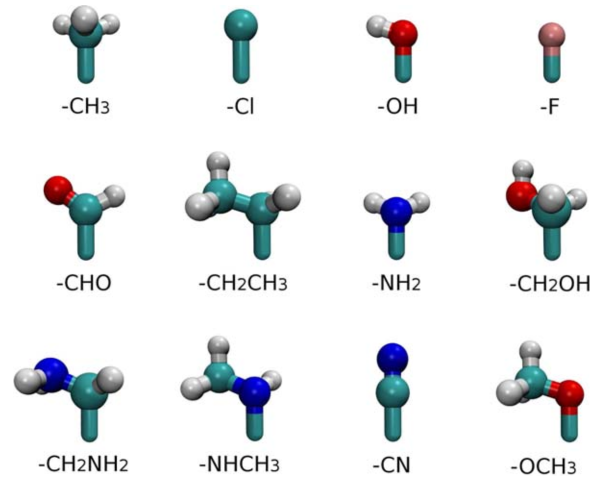

FIGURE 1 Structures of the 12 functional groups studied in this work.

sampled every 30 degrees unless rotational symmetry is exploited, *e.g.*, rotation of CH_3 is only sampled at 0, 30, 60, 90 degree. Reference mFEP calculations are performed with ten 5-ns windows, that is, $\lambda{=}0, 0.1, \cdots, 0.9$ in the forward direction (benzene to its derivative) and $\lambda{=}1.0, 0.9, \cdots, 0.1$ in the backward direction (the derivative to benzene). With 12 functional groups studied via both forward and backward mFEP, there are 240 ($12 \times 10 \times 2$) 5-ns mFEP trajectories in total. For comparison, we also perform altogether 24 (12×2) 50-ns sFEP calculations in both directions using NAMD. Different from mFEP, sFEP involves MD simulation at a single end state, i.e., $\lambda = 0$ or $\lambda = 1$. Errors of VSS and sFEP calculations are estimated via the subsampling analysis.[26,53,55]

Our second test system consists of T4 Lysozyme in complex with a benzene ligand. The system is constructed with the protein (pdb code 187L) solvated in water and neutralized under 0.15*M* NaCl. As for the benzene derivative, since not all 12 ligands can bind to the nonpolar pocket inside the T4 Lysozyme,[60] we choose three binders with different sizes (toluene, ethylbenzene, and p-xylene) and one nonbinder (phenol). VSS calculations are performed based on a single 25-ns trajectory of a benzene-Lysozyme complex, where data from all 6 sites of benzene are combined based on the chemical equivalence to improve the sampling efficiency. The corresponding mFEP calculations employ 20 10-ns windows, except for the case of toluene, where 2.5-ns for each window is found to be sufficient. Following Mobley et al.,[61] Val 111 is constrained in all Lysozyme simulations to the holo position (pdb code 187L). A summary of all simulations in this work is given in Table I.

All NAMD simulations reported here are performed with the 2.9 release of NAMD[51] and the CHARMM general force field for small molecules.[62] A timestep of 2 fs is adopted, with bonds involving hydrogen atoms constrained using RATTLE[63]

Table I Summary of Simulations Reported in This Work

Test System	Simulation Type	Trajectory Length	No. Trajectories
Benzene	NAMD mFEP	5 (ns)	240
	NAMD sFEP	50 (ns)	24
	VSS	50 (ns)	1
Benzene-T4 Lysozyme complex	NAMD mFEP	10 or 2.5 (ns)	160
	VSS	25 (ns)	1

and water geometries maintained using SETTLE.[64] The multiple-time-stepping algorithm is used, with short-range forces calculated every step and long-range electrostatics calculated every 2 steps. The cutoff for short-range non-bonded interactions is set to 12 Å, with a switching distance of 10 Å. Assuming periodic boundary conditions, the PME method[58] with a grid density of at least 1/Å^3 is employed for computation of long-range electrostatic forces. Langevin dynamics with a damping coefficient of 1 ps^{-1} is used to keep the temperature constant at 300 K, while a Nosé-Hoover-Langevin piston[65] is used to keep the pressure constant at 1 atm. The ParamChem program[66,67] and CGenFF[62] are used for parameterization of molecules. Parameters of the ligand benzylamine (-CH_2NH_2 substitution) is further optimized using the VMD FFTK plugin[30] and the program Gaussian.[68]

RESULTS AND DISCUSSION

Benzene in Water

We begin with a relatively simple test system, namely the mutation of a benzene into its derivative in water. The derivatives are constructed by substituting an aryl hydrogen by 12 different functional groups shown in Figure 1. The free energy change of the alchemical transformation is carried out in solution via VSS, NAMD sFEP, and NAMD mFEP calculations. The forward (benzene to its derivative) and backward results are combined using Simple Overlap Sampling (SOS)[69] implemented in ParseFEP plugin of VMD[32] and summarized in Table II. The convergence of mFEP calculation is reflected in the agreement between forward and backward calculations, cf. Figure 2, as well as the small error obtained from SOS analysis ($\leq$ 0.05 kcal/mol). Therefore, the mFEP SOS results are chosen as reference to be compared with NAMD sFEP and VSS. This

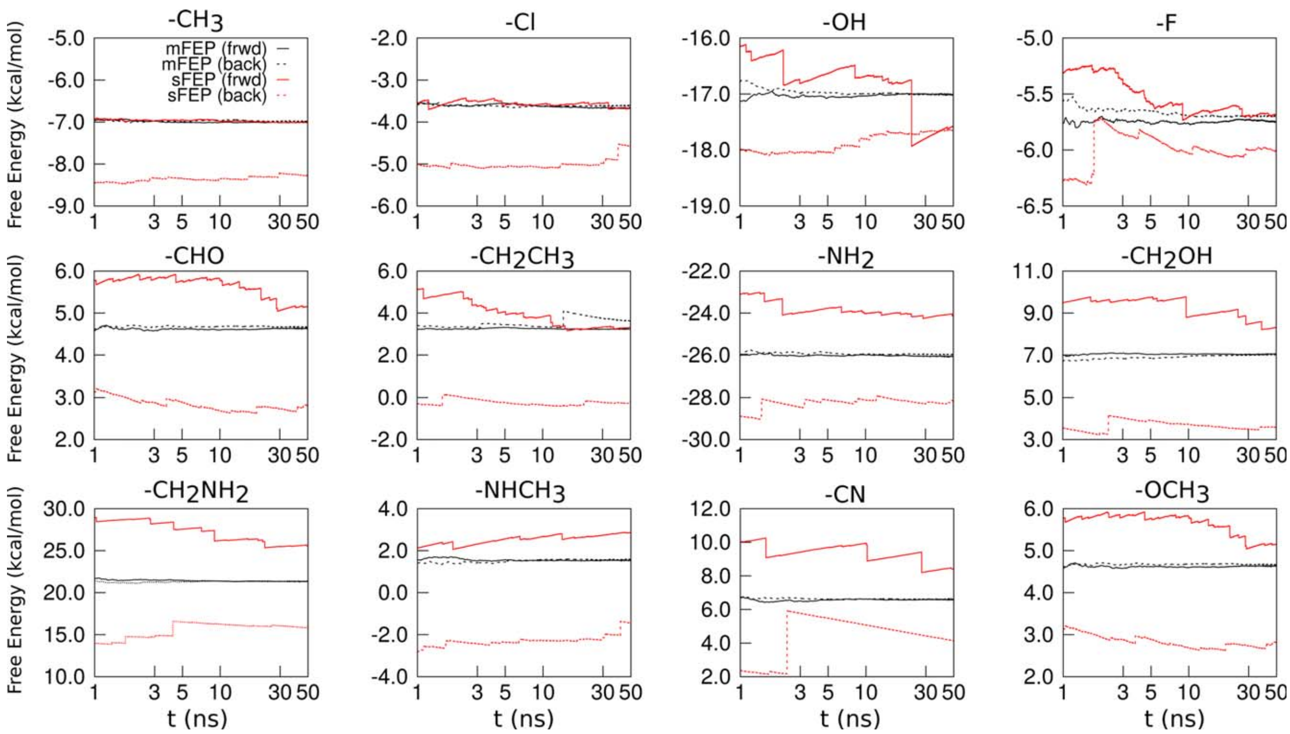

FIGURE 2 NAMD mFEP (black) and sFEP (red) results of the free energy change upon substituting an aryl hydrogen by the 12 functional groups in solution. Results from forward (frwd) and backward (back) calculations are shown in solid and dashed lines, respectively. For NAMD mFEP calculations, t represents the total simulation time, *i.e.*, each mFEP window is simulated for t/W, where W represents the number of windows.

Table II Free Energy Change Upon Substituting a Hydrogen of Benzene in Dolution by the 12 Functional Groups Shown in Figure 1

Substitution	mFEP+SOS	sFEP+SOS	VSS+SOS	sFEP	VSS
$-CH_3$	-7.00 ± 0.02	-6.99 ± 0.01	-7.03 ± 0.01	-7.01 ± 0.05	-7.00 ± 0.05
-Cl	-3.58 ± 0.03	-3.64 ± 0.02	-3.65 ± 0.01	-3.67 ± 0.42	-3.67 ± 0.05
-OH	-17.01 ± 0.02	-16.99 ± 0.01	-17.01 ± 0.01	-17.57 ± 0.37	-16.98 ± 0.23
-F	-5.69 ± 0.02	-5.72 ± 0.01	-5.90 ± 0.01	-5.68 ± 0.14	-5.87 ± 0.07
-CHO	4.64 ± 0.03	4.70 ± 0.03	4.59 ± 0.02	5.16 ± 0.32	4.46 ± 0.49
$-CH_2CH_3$	3.26 ± 0.03	3.33 ± 0.06	3.21 ± 0.01	3.32 ± 0.33	3.42 ± 0.12
$-NH_2$	-26.03 ± 0.04	-26.05 ± 0.05	-26.02 ± 0.01	-24.12 ± 0.35	-25.83 ± 0.18
$-CH_2OH$	7.04 ± 0.03	7.10 ± 0.09	7.12 ± 0.02	8.31 ± 0.82	7.49 ± 0.39
$-CH_2NH_2$	21.40 ± 0.05	21.59 ± 0.16	21.25 ± 0.04	25.60 ± 0.79	22.01 ± 0.34
$-NHCH_3$	1.61 ± 0.04	1.65 ± 0.04	1.87 ± 0.02	2.87 ± 0.34	2.41 ± 0.34
-CN	6.66 ± 0.04	6.26 ± 0.20	6.46 ± 0.09	8.43 ± 1.25	5.55 ± 1.15
$-OCH_3$	4.55 ± 0.04	4.64 ± 0.22	4.99 ± 0.10	6.29 ± 1.34	7.48 ± 0.26

Forward and backward calculations are combined using SOS analysis[70] implemented in ParseFEP.[32] The "sFEP" column refers to forward NAMD sFEP calculation. All values are in the unit of kcal/mol.

type of comparison, viz. comparing data of a single leg in the thermodynamic cycle, allows an unambiguous analysis for the performance of the latter two methods.

As shown in Table II, when both forward and backward calculations are performed and combined via SOS analysis to yield a single free energy change, mFEP, sFEP, and VSS all produce comparable results. Here, 12 additional "vanilla" trajectories containing benzene derivatives are collected in order to perform the backward VSS calculations for the corresponding SOS analysis. Combining FEP calculations from both directions is not what VSS is designed for, although the SOS analysis provides an additional reference that the plugin indeed functions as expected. The true power of VSS lies in post-processing a single trajectory for a quick FEP scan, based on the sampling of a single end state. By doing so, one saves computing resources spent on collecting trajectory for the same reference state. For instance, VSS calculation of the -OH substitution took less than two hours using four i7-4770 cores on a desktop, which is 20 times faster than the corresponding mFEP calculation performed on the same machine. The former calculation is based on scanning 12 different dihedral conformations and its speedup will further increase if zero (-Cl) or a reduced number ($-CH_3$, exploring symmetry) of dihedrals needs to be sampled. Clearly, if VSS could yield reasonably reliable free energy estimates, it would be a rather appealing toolkit for a quick, first scan for multiple substitutions. So how reliable are VSS calculations based on a single, vanilla trajectory?

As listed in the last column of Table II, VSS results show an overall good agreement with the mFEP + SOS reference: the difference between the two types of calculations is smaller than 0.5 kcal/mol for seven functional groups, $-CH_3$, -Cl, -OH, -F, $-CH_2CH_3$, $-NH_2$, $-CH_2OH$, and is between 0.5 and 1.0 kcal/mol for two other groups, $-CH_2NH_2$ and $-NHCH_3$. Notably, the former list contains functional groups with different polarity, e.g., -OH and $-CH_3$, while the latter list consists of somewhat sizable groups, such as $-CH_2OH$ and $-CH_2NH_2$. Finally, VSS calculations of -CHO and -CN did not converge based on a 50-ns trajectory, where the former can be considered a border-line case. This indicated by their continuously decreasing curves shown in Figure 3 as well as the large errors obtained from subsampling analysis (Table II). VSS results obtained using a 200 ns trajectory for these two functional groups (Supporting Information Figure S1), along with the special case of $-OCH_3$, are discussed later.

Figure 2 indicates that results of the forward NAMD sFEP calculation agree better with the mFEP references than the corresponding backward calculations. This result echoes previous findings[70] on the asymmetry of FEP calculations performed in the two directions. Detailed error analsyis of the sFEP calculations following the protocols of Lu and Kofke[71] is provided in the Supporting Information (Figure S2). In contrast to VSS, results of NAMD sFEP are within 0.6 kcal/mol of the mFEP + SOS references only for the first 6 examples, $-CH_3$, -Cl, -OH, -F, -CHO, $-CH_2CH_3$, as shown in Table II. While NAMD sFEP and VSS produce generally similar data, notable difference (> 1 kcal/mol) between the two methods are found in $-NH_2$, $-CH_2NH_2$,-CN, $-OCH_3$. With the exception of $-OCH_3$, VSS outperforms NAMD sFEP for the remaining three functional groups and additionally for $-CH_2OH$ and $-NHCH_3$, suggesting that more efficient sampling is achieved in VSS.

Comparison of NAMD sFEP and VSS

From the implementation perspective, VSS differs from NAMD sFEP in a number of ways. First, dihedrals in

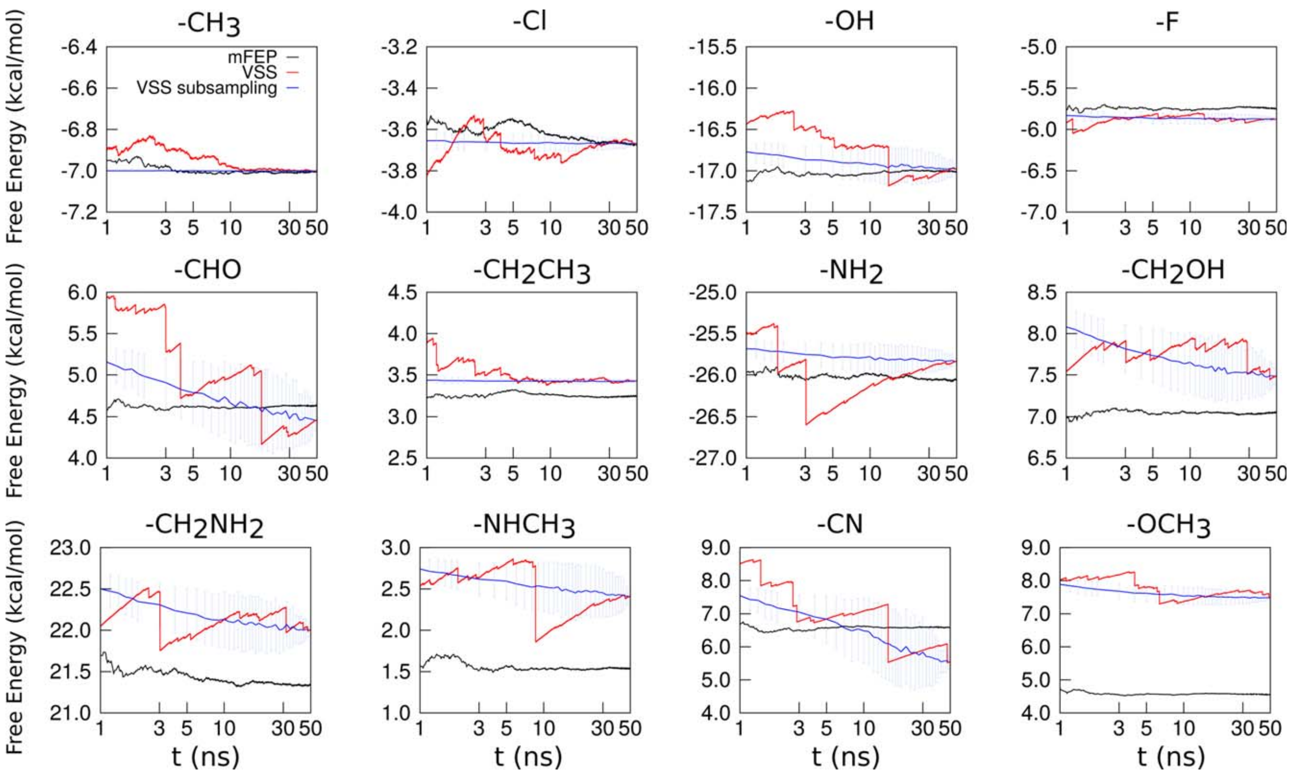

FIGURE 3 Free energy change upon substituting an aryl hydrogen of benzene by the 12 functional groups in solution. Results of forward mFEP and VSS are colored in black and red, respectively, while subsampling analysis results based on the latter calculation is colored in blue.

polyatomic functional groups are sampled "explicitly" in VSS, i.e., each desired dihedral is sampled at equal spacing and the resulting $U_2^{bonded}(\mathbf{r_2})$ is used to evaluate Eq. (7). In single-step FEP via NAMD, such sampling is performed "implicitly," i.e., L_2 evolves freely according to $U_2^{bonded}(\mathbf{r_2})$ in the reference-state simulation. For a given frame, any selected dihedral can only adopt a single value in NAMD sFEP, whereas multiple values can be sampled in VSS through explicit rotation of the dihedral (see Figure 4a). The efficiency of the latter calculation is thus improved. Without such rotation, the VSS calculation may significantly deviate from the mFEP result. Take -CH_2OH for example. Fixing the dihedral angle during the VSS calculation at 0°, 120°, and 240° leads to a result of 11.69, 8.50, and 7.93 kcal/mol, respectively (Figure 4b). Compared with the mFEP + SOS reference (7.04 kcal/mol), these results show deviations up to 4.65 kcal/mol, whereas with dihedral rotation enabled, the deviation is only 0.45 kcal/mol. Therefore, the sampling of dihedral rotations is indispensable for certain functional groups.

Second, VSS differs from NAMD sFEP in the treatment of L_2 atoms that are not part of the substituting functional group. In the former calculation, on-site mutation scheme is used: L_2 atoms are assigned the coordinates of their counterparts in L_1, regardless of any difference in atomic charges brought by their different substituting functional groups. As a result, VSS samples the configurational space efficiently for substitutions that perturb the charges of L1 atoms, such as -NH_2, -CH_2NH_2, and -CN. Take -NH_2 as an example. Upon the substitution, charges of all carbons on the benzene ring are altered, except for C5 at para-position of -NH_2, see Figure 4c. This leaves C5 and H5 the only two common atoms of L_1 (benzene) and L_2 (aniline). Consequently, in NAMD sFEP calculations, L_2 can rotate freely relative to L_1 around the C5-H5 axis. Such free rotation is purely a consequence of employing the dual-topology during FEP calculations. Yet, it creates a sampling issue similar to the "wandering ligand" problem studied extensively in the calculation of absolute binding free energy (see Ref. 17 and references therein). Compared with mFEP, this issue is particularly relevant for sFEP calculation, where sampling is already more challenging. To confirm that the rotation of the benzene ring is the cause of the problem, we create a fake aniline molecule with slightly modified atomic charges, thereby, eliminating the rotation described above. Subsequently, the difference between NAMD sFEP and mFEP + SOS results immediately reduces

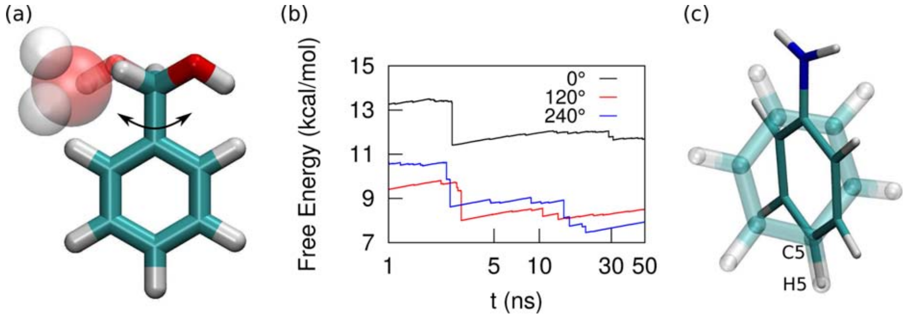

FIGURE 4 Explicit sampling of dihedrals in VSS. (a) Rotation of the C-C-C-O dihedral in VSS calculation for -CH_2OH. (b) VSS results of -CH_2OH without sampling the dihedral. The three curves correspond to results with the aforementioned dihedral adopting a single value. (c) Sampling issues in NAMD sFEP due to molecule rotation. Upon the mutation of benzene (L_1) to aniline (L_2), atomic charges of five carbons on the ring are altered, allowing the rotation of L_2 relative to L_1 under a dual-topology setup.

from ~2 kcal/mol to less than 0.3 kcal/mol. Therefore, for such functional groups, restraints are highly desirable during NAMD sFEP calculations, in order to ensure that L_2 does not drift away from the position and orientation of L_1.

The final difference between VSS and NAMD sFEP is the treatment of the substituting functional group itself—while this group is treated as a rigid unit in VSS, it can evolve freely in NAMD sFEP calculation. Based on the comparison of VSS and mFEP + SOS results, the effect of such difference appears to be small for most functional groups. The -OCH_3 group represents a notable exception and will be discussed in the following section.

Accuracy of VSS Via Subsampling Method

Estimating convergence and the error for a sFEP calculation is nontrivial. For this purpose, VSS employs the subsampling method,[26,53–55] which produces monotonic and smooth curves, instead of saw-tooth curves typically observed in sFEP results.[25] The method calculates the averaged free energy change (ΔA_n) based on a block-averaging procedure. Briefly, subsets (blocks) are first created by randomly selecting n data points without replacement from the total N data points. The process is repeated until the desired number of blocks m are obtained ($m \approx 100\times N/n$).[55] A block free energy $\Delta A_{n,j}$ is then obtained as,[54,55]

$$\Delta A_{n,j} = -\frac{1}{\beta}\ln\left(\frac{1}{n}\sum_{i\in block\, j} e^{-\beta u_i}\right), \tag{9}$$

where j is the block index and $n = 1,2,.,N$. We note that u used in the above equation is already weighted by $e^{-\beta U_2^{bonded}(\mathbf{r}_2)}$. The subsampled free energy ΔA_n is then obtained as the average of $\Delta A_{n,j}$ from all blocks,

$$\Delta A_n = \frac{1}{m}\sum_{j=1}^{m}\Delta A_{n,j}\,, \tag{10}$$

with a standard deviation[55]

$$\sigma_n = \sqrt{\frac{1}{m}\sum_{j=1}^{m}\left(\Delta A_{n,j} - \Delta A_n\right)^2}\,, \tag{11}$$

which gives the unsigned error in Figure 3 via[54] $2\sigma_n/\sqrt{N/n}$. Clearly, the above error estimation is valid only when the blocks are uncorrelated (independent). The number of independent blocks decreases when n approaches N. In particular, when $n = N$, all blocks are simply permutation of the original data set, resulting in $\sigma_N = 0$. Therefore, the correlation of the blocks results in an artificial drop in the estimated error as n increases, see e.g., the error bars in Figure 3. For this reason, we take the maximum error recorded for all $n = 1,2,\ldots,N$ to characterize the uncertainty of a VSS calculation.

The free energy obtained from VSS and subsampling on VSS data are depicted in Figure 3. For comparison, the mFEP forward simulation is also shown. The subsampling ΔA_n result (blue curve) is a smoothly decreasing curve and, therefore, provides a better convergence check than the original, sawtooth-like curve. As shown by these results, VSS calculations for -CHO and -CN are not yet converged, as their subsampling curves continue to decrease even at 50 ns. Interestingly, the subsampling method recognizes that the corresponding VSS calculation of -OCH_3 is converged with an error of 0.26 kcal/mol. However, the final VSS result for -OCH_3 is 2.93 kcal/mol away from the mFEP + SOS reference. In comparison, the subsampling method finds a much larger error of 1.34 kcal/mol for the corresponding NAMD sFEP calculation, the result of which deviates from the mFEP + SOS

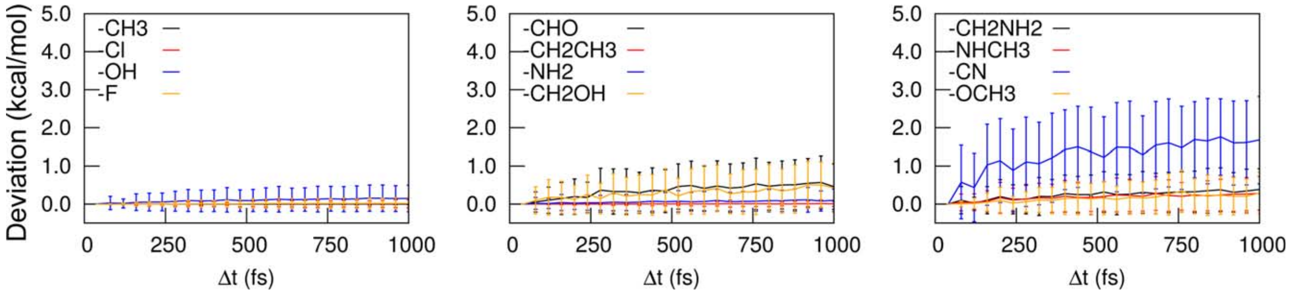

FIGURE 5 Magnitude of deviation respect to changing the trajectory output time interval Δt for VSS calculation. VSS results shown in Table II are taken as the reference. Results obtained from a given output frequency (e.g. $\{0, 80, \cdots\}$ and $\{40, 120, \cdots\}$) are averaged, and the standard deviations are shown as the error bar.

reference by 1.74 kcal/mol. These results together suggest that unlike NAMD sFEP calculation of -OCH_3, where convergence is an issue, the error of VSS is likely of a different nature. Indeed, our preliminary analysis shows that sampling of the C-O-C angle (one of the carbons comes from the benzene ring) is particularly important for this functional group. We plan to enable angle vibration in future versions of VSS and further examine its performance on the -OCH_3 group.

It is worth mentioning that the subsampling data can be used to estimate the upper bound of free energy change, even if the calculation is not converged. As shown by Zuckerman and Woolf,[26] the expected value of a forward FEP calculation and the exact ΔA are related via

$$\Delta \mathbb{A}_n \geq \Delta \mathbb{A}_{n+1} \geq \ldots \geq \Delta A_{exact} \tag{12}$$

where $\Delta \mathbb{A}_n$ is obtained from Eq. (10) with $N, m \rightarrow \infty$. In practice, one only has access to ΔA_n, which is evaluated with finite N and m. Its uncertainty largely determines the reliability of the estimated upper bound. Hence, we use $\Delta A_n + 2\sigma_n/\sqrt{N/n}$, i.e. the upper limit of the error bar, as an estimator to the upper bound of ΔA_{exact}. Note that the estimation is reliable only before the error bar starts to decrease again due to the correlation of blocks described earlier. For instance, the last reliable upper bound estimation appears at 20-ns for -CHO and -CN, with the values of 5.06 and 6.90 kcal/mol, respectively. These are decent upper bounds of ΔA_{exact}, considering their corresponding mFEP + SOS results are 4.64 and 6.66 kcal/mol, respectively. Extending the 50-ns VSS calculations to 200 ns yields a converged result of 4.56 ± 0.32 for -CHO and a nearly converged result 6.26 ± 0.73 kcal/mol for -CN (Supporting Information Figure S1). However, given the much increased computational cost, a user may find it more desirable to take the estimated upper bound from the shorter VSS run and pursue mFEP calculations if needed.

Before proceeding to the next section, we shall explain the impact of trajectory length and output frequency on VSS result. The reported VSS calculations are carried out using a 50-ns trajectory with an output frequency of 40 fs, the latter of which is chosen to be sufficiently larger than the correlation time of $\exp(-\beta u)$ (Supporting Information Table S1). When a shorter trajectory is used, the system may suffer from insufficient exploration of the configurational space, resulting in an unreliable free energy estimate. As shown in Supporting Information Figure S3, for very small functional groups, such as -CH_3, -Cl, -OH, -F, simulations up to 20 ns are sufficient to provide a reliable free energy estimate, while other groups tend to require a longer trajectory, i.e., more than 40 ns. When a trajectory is saved less frequently, important configurations, which contribute dominantly in Eq. (7), may be missed due to a larger output time interval (Δt). The effect of Δt is estimated through resampling the 50-ns VSS data, based on different output time intervals ranging from 40 fs to 1000 fs. Results obtained with the same Δt (e.g. $\{0, 80, \cdots\}$ and $\{40, 120, \cdots\}$) are averaged, and their standard deviations are shown as the error bars in Figure 5. The absolute deviation of the above data from the VSS result obtained with $\Delta t = 40$ fs (Table II) are depicted in Figure 5. As expected, VSS calculations with lower trajectory output frequency (larger Δt) yields a larger deviation. Generally, a small substitution such as -Cl, -F, -OH, -NH_2, -CH_3 are not affected much by a less frequent trajectory output, while a larger substitution like -OCH_3, -$NHCH_3$, -CH_2OH, -CH_2NH_2, can have deviations up to ~1.0 kcal/mol. Likely due to its size and extended geometry, -CN suffers the most from increased Δt: the deviation quickly grows to ~3.0 kcal/mol. This agrees with the aforementioned result that it takes at least 200 ns for VSS to yield a converged result for -CN. Finally, we mention that while a frequent output may be desirable to avoid missing statistically significant data points, one may always use a low output frequency combined with a long simulation to obtain sufficient statistics.

It is worth adding that while only tests on aromatic ligands are discussed above, VSS also supports the free energy scan for

Table III **Binding Free Energy Change to T4 Lysozyme Upon Alchemically Transforming a Benzene Into Its Derivatives**

Ligands	mFEP+SOS	mFEP FWD	mFEP BWD	VSS
phenol (-OH)	-15.10 ± 0.07	-14.87 ± 0.43	-15.31 ± 1.27	-14.29 ± 0.04
toluene ($-CH_3$)	-7.89 ± 0.02	-7.91 ± 0.30	-7.86 ± 0.33	-7.87 ± 0.28
ethylbenzene ($-CH_2CH_3$)	2.44 ± 0.02	2.06 ± 0.42	2.89 ± 0.23	1.46 ± 0.42
p-xylene ($-CH_3$)	-13.21 ± 0.04	-13.24 ± 0.72	-13.20 ± 0.64	-14.89 ± 0.48

Both forward (FWD) and backward (BWD) free energy changes are listed, together with their combined result (SOS). Simulations are performed with Val 111 side chain confined (see text for details). All values are in the unit of kcal/mol.

aliphatic ligands. While the latter compounds are more flexible, we obtain a reliable free energy estimation when alchemically transforming an n-hexane into an n-heptane (VSS: 0.44 ± 0.10 kcal/mol, mFEP + SOS: 0.53 ± 0.02) and an n-hexane into a 3-methylhexane (VSS: -6.08 ± 0.10 kcal/mol, mFEP + SOS: -6.06 ± 0.03), both in solution.

T4 Lysozyme with Benzene

Free energy calculation involving proteins is generally expected to be challenging, as sampling of protein DOFs can significantly affect the calculation result. For instance, the dihedral rotation of side chain of Val 111 is known to cause convergence issues in free energy calculations involving T4 Lysozyme.[61] Indeed, we notice that without any constraint on Val 111, the forward and backward mFEP calculations on T4 Lysozyme yield inconsistent results. Therefore, we follow the confine-and-release treatment from Mobley, et al.[71] and constrain the dihedral angle of Val 111 in its *holo* configuration (pdb code 187L). With this constraint, mFEP calculations from both directions yield nearly identical results. Four ligands are then selected for VSS calculation: phenol, toluene, ethylbenzene, and p-xylene. As a polar ligand, phenol does not bind to the nonpolar pocket of T4 Lysozyme, while the other three ligands form complexes with the protein.[60] Except for toluene (50-ns simulation), all mFEP simulations are 200-ns long, while VSS utilizes a single 25-ns MD trajectory. Exploiting the chemical equivalence of the 6 carbons on benzene, we are able to obtain converged results via combining data from substitution on different carbons. The benchmark of the mFEP calculation is 0.028 days/ns using 128 cores on a high performance cluster, while the VSS calculation took approximately 8 hours using four cores on a desktop.

As shown in Table III, for toluene and p-xylene, the free energy changes obtained by forward and backward mFEP calculations are almost identical. In contrast, the mFEP results of phenol and ethylbenzene in the two directions are off by 0.44 and 0.83 kcal/mol, respectively, revealing the difficulty in sampling when the molecule is placed in a protein. Since VSS is performed under the forward direction, we will compare its performance to both the forward mFEP and the mFEP + SOS results.

Most VSS results follow closely their forward mFEP reference with a deviation up to 0.6 kcal/mol. For toluene, which rotates freely inside the binding pocket as benzene does, the VSS result agrees almost perfectly with that of mFEP. Compared with benzene and toluene, ethylbenzene and p-xylene rotate less freely inside the pocket. Yet the deviation of VSS data from forward mFEP is less severe in ethylbenzene (0.60 kcal/mol) than in p-xylene (1.65 kcal/mol). In order to better understand this deviation, we perform clustering analysis on the first window of the backward mFEP trajectory for both ligands, i.e., when they are still fully coupled to the environment. The geometry of major clusters, which represent more than 50% of the trajectories, are depicted in Figure 6. As a reference, major clusters of the benzene-Lysozyme simulation used by VSS calculation are shown in red. Comparison with the crystal structure reveals that the loop next to the binding

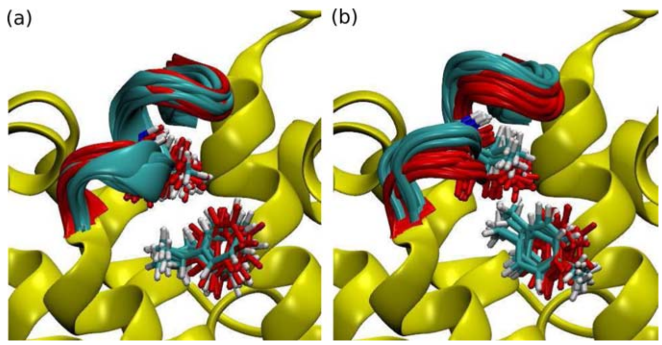

FIGURE 6 Clustering analysis of trajectories used in mFEP and VSS calculations for T4 lysozyme. Structures of the loop (resid 106 to 114) and ethylbenzene from major clusters are depicted. The confined residue Val 111 is also depicted (see text). (a) Clusters of ethylbenzene-Lysozyme complex. Structures of major clusters (resid 106 to 114) and ethylbenzene are depicted in cyan. For comparison, clusters of benzene-Lysozyme complex are depicted in red. (b) Clusters of p-xylene-Lysozyme complex (cyan). The protein undergoes a conformational change in p-xylene mFEP simulation.

pocket (resid 106 to 114) maintains its conformation in both benzene and ethylbenzene trajectories, while it shifts away from p-xylene during the mFEP simulation, i.e., the loop conformation sampled in p-xylene mFEP differs from that sampled in the trajectory processed by VSS. The same loop also changes its conformation when the protein "binds" to phenol, although such change is much smaller.

In addition to protein conformational changes, the clustering analysis also reveals that ethylbenzene and p-xylene each has its preferred orientation: the ethyl group of the former points away from Val 111, while one methyl group of the latter points towards it (Figure 6). While VSS considers all possible orientations to achieve the best sampling efficiency, out of curiosity, we post-process our data to study the effect of performing the calculation with a single, preferred orientation: we first align the protein structure from the benzene-lysozyme trajectory to the largest cluster obtained from ethylbenzene or p-xylene mFEP simulation; then a mutation site is selected on the benzene ring such that the resulting functional group best aligns with the mFEP reference. Following this protocol, we obtain 2.75 and -14.99 kcal/mol for ethylbenzene and p-xylene, respectively. It is worth mentioning that the above results can be highly sensitive to the adopted protocol, *i.e.*, if a slightly different reference orientation or alignment scheme is used, the outcome tends to vary.

While VSS results follow closely those from the forward mFEP calculation, ideally they should be comparable with the mFEP + SOS references. Table III indicates that VSS generally deviates from the mFEP + SOS references by 1 kcal/mol. When combining the data from the protein-ligand complex simulation and the free ligand simulation to yield the relative binding free energy, we find that VSS separates the nonbinder from the binders successfully. In addition, it provides acceptable estimation of relative binding free energy (bias around 1 kcal/mol) for most ligands, except for p-xylene where a protein conformational change takes place as described above. Note that the reported values here are obtained under the Val 111 dihedral confinement. To recover data that are comparable with experiments, one needs to include the free energy contribution of releasing the Val 111 dihedral constraint.[61,71] Since such calculation contributes equally to mFEP and VSS results, it does not affect our evaluation of the performance of the latter.

CONCLUSIONS

In this study, we report virtual substitution scan, a toolkit designed to post-process a vanilla MD trajectory to compute the relative free energy change upon substituting aryl hydrogens by small functional groups. Unlike previous methods, VSS utilizes a trajectory with no soft-core potential and explicitly samples dihedral rotations with an on-site mutation scheme. It supports the recycling of trajectories initially not designed for free energy calculations. Based on a single, 50-ns trajectory of benzene in water, we employed VSS to compute the free energy change upon substituting aryl hydrogens by 12 small functional groups. The calculation for a given functional group takes less than two hours on a 4-core desktop, which is ~20 times faster than the corresponding mFEP calculation. Results obtained with VSS are reasonably accurate (error < 1 kcal/mol) for 9 of the 12 tested substituting functional groups, including relatively sizable and polar ones. For the remaining 3 groups, subsampling analysis suggests that two of the calculations (-CHO and -CN) are not converged, although a decent estimate for the free energy upper bound can be extracted through the analysis. The only ligand that shows an obvious systematic bias is $-OCH_3$. This bias is likely caused by neglecting angle sampling in the current VSS implementation and will be further examined in our future work. The performance of VSS is also tested by computing the relative binding free energy of four benzene derivatives to the T4 Lysozyme. Based on a 25-ns trajectory, VSS produced the correct trend to separate binders from the non-binder, and its deviation from the mFEP reference obtained with up to 200-ns simulation is generally around 1 kcal/mol. The unavoidable limitation of VSS, as a type of single-step FEP calculation, lies in the physical size of the functional group, *e.g.*, -CHO and -CN, as well as protein conformational changes that are not captured by the input trajectory. Additionally, it will likely fail if the functional group carries a net charge. However, for small groups frequently encountered in protein-ligand binding and lead optimization studies, VSS can provide a rapid, first scan of multiple substitutions based on a single, vanilla MD trajectory.

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